

# Web Technology



## React – AJAX, SPA Components and Props

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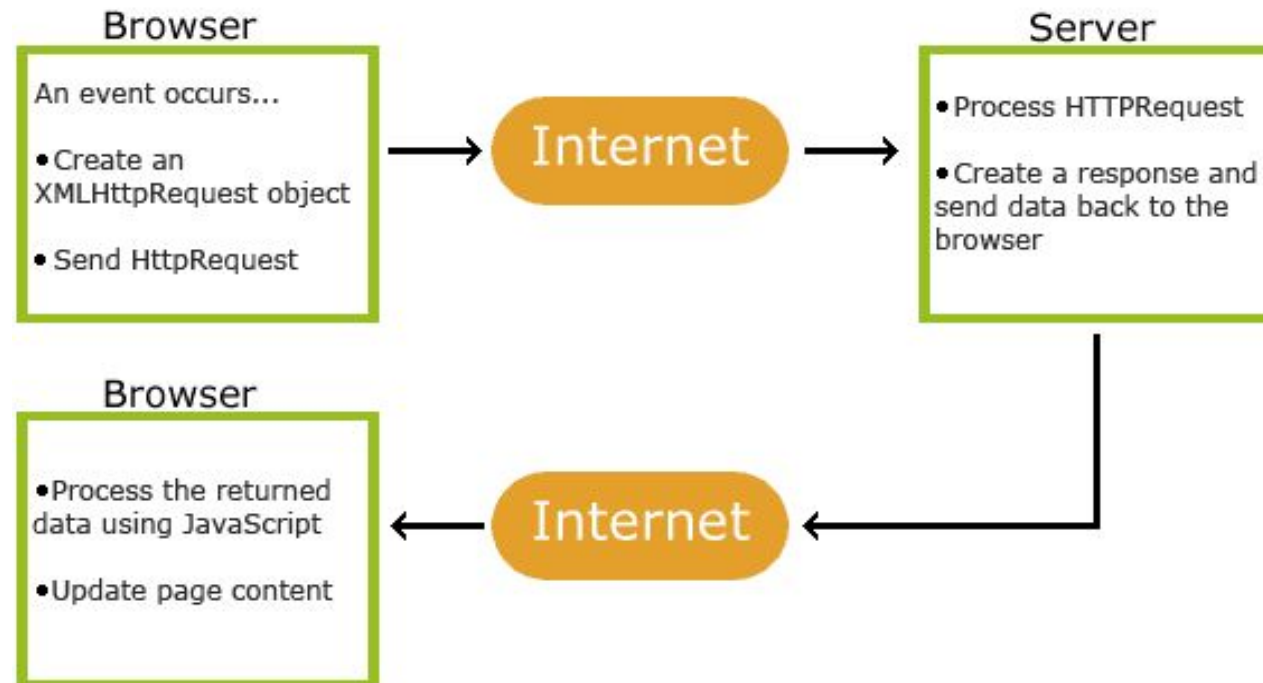
- **Not a programming language**
- **Technique for Accessing Web Servers and Web Pages**
- **Asynchronous JavaScript and XML**
- **Utilizes JSON instead of XML**

**DOM manipulation** → changes part of the page on the client side.

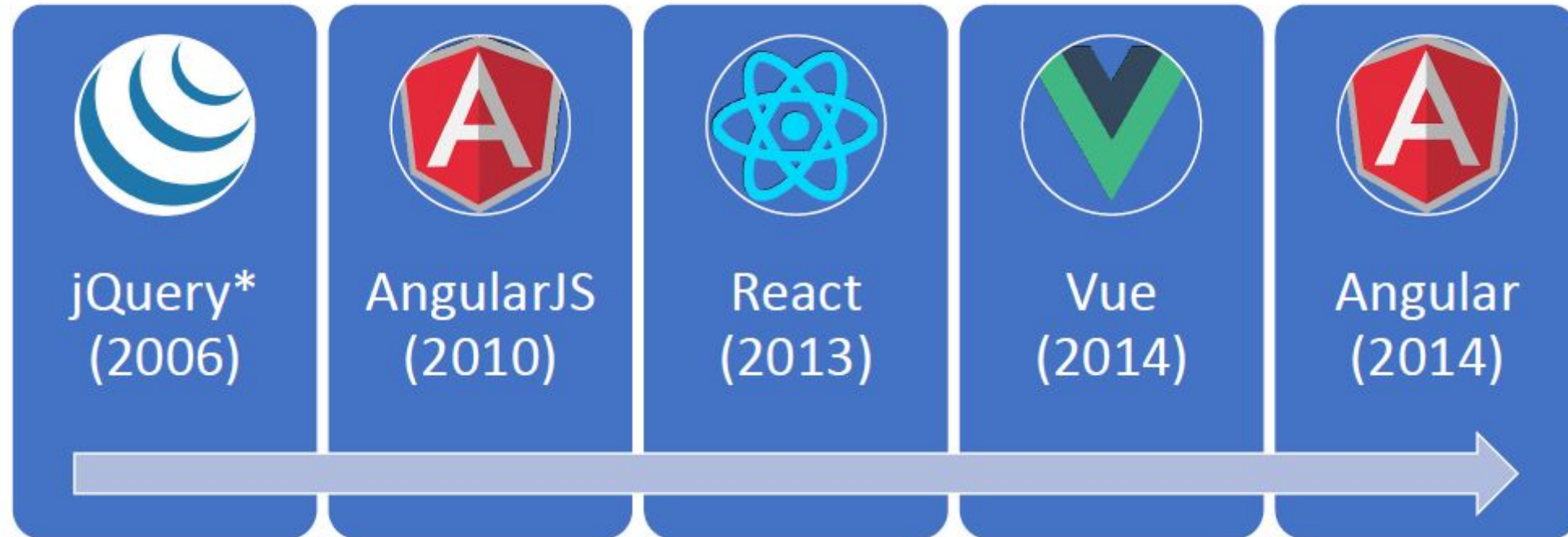
**AJAX** → gets new data from the server without reloading the page.

# AJAX

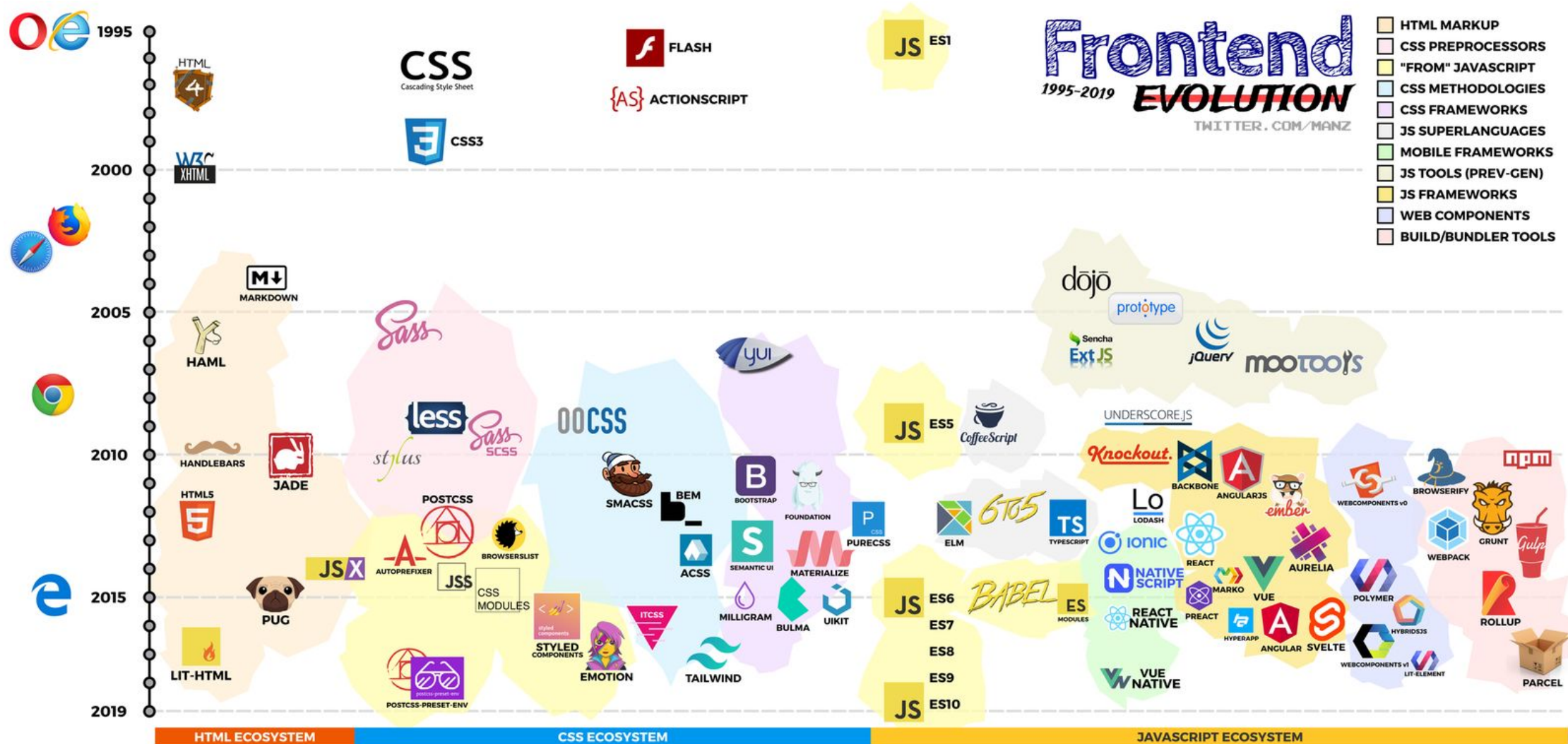
- **Not a programming language**
- **Technique for Accessing Web Servers and Web Pages**
- **Asynchronous JavaScript and XML**
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# Front End Timeline



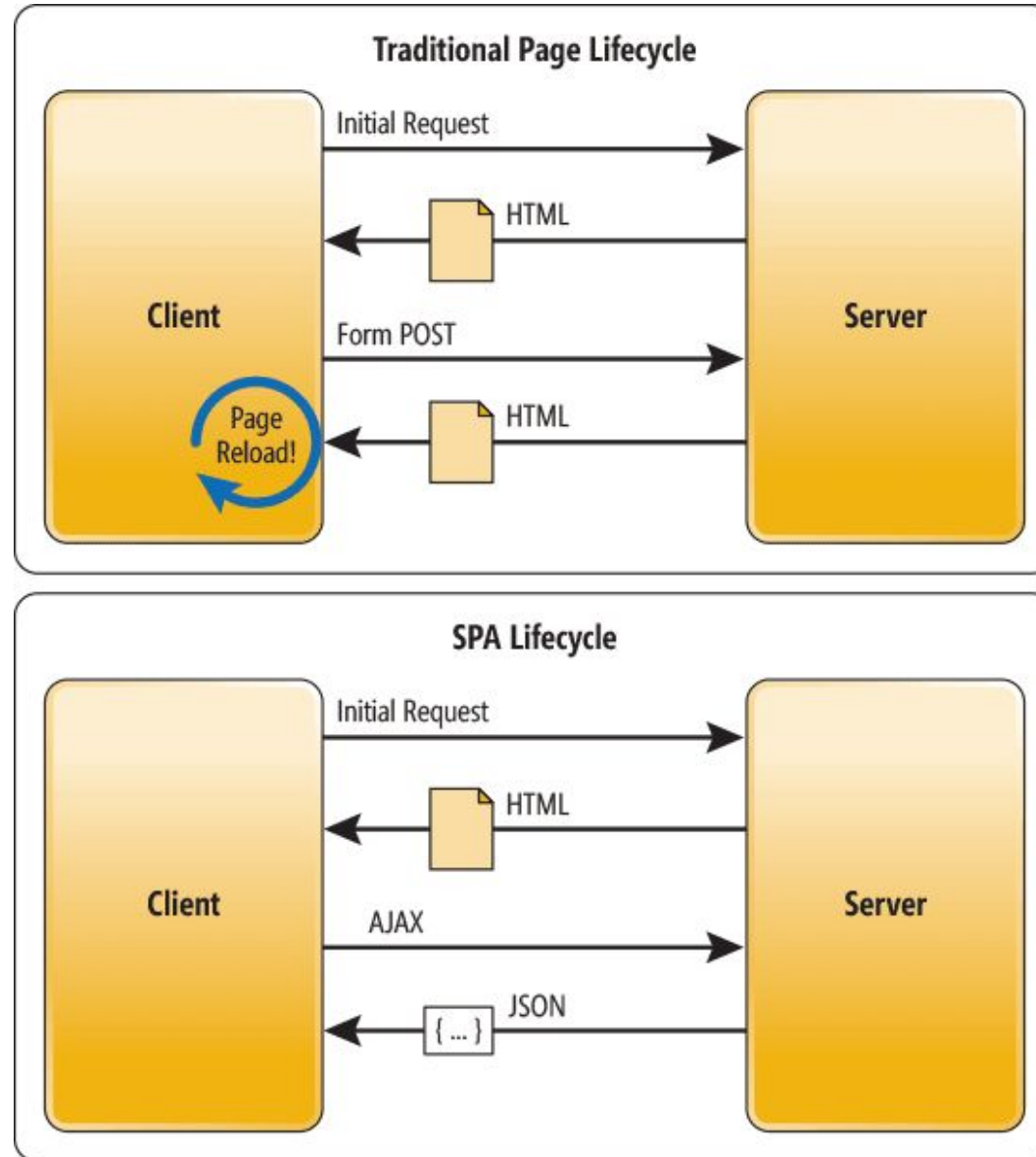
# Front End Timeline



# Single Page Application (SPA)

- **Single HTML file**
- **Dynamic content updates**
- **Component-based structure**
- **Client-side routing**

# Single Page Application (SPA)





# Traditional vs React



Traditional  
approach



React  
approach

## Project Structure Comparison

### Traditional Website

```
Project
|
├─ index.html
├─ style.css
└─ script.js
```

### React

```
src/
|
├─ App.jsx
├─ App.css
├─ Home.jsx
├─ Home.css
└─ main.jsx
```



# Front End Development Tasks

|               |                                               |
|---------------|-----------------------------------------------|
| App state     | Data definition, organization, and storage    |
| User actions  | Event handlers respond to user actions        |
| Templates     | Design and render HTML templates              |
| Routing       | Resolve URLs                                  |
| Data fetching | Interact with server(s) through APIs and AJAX |

# Running React

- **Install Node.js**

# Running React

- **Install Node.js**
  - [Download and Install - Node.js — Download Node.js®](#)
- **Open a Terminal**
  - **`npx create-react-app app-name`**
  - **Change directory/folder to `app-name`**
  - **`npm start`**
- **Code in App.js**

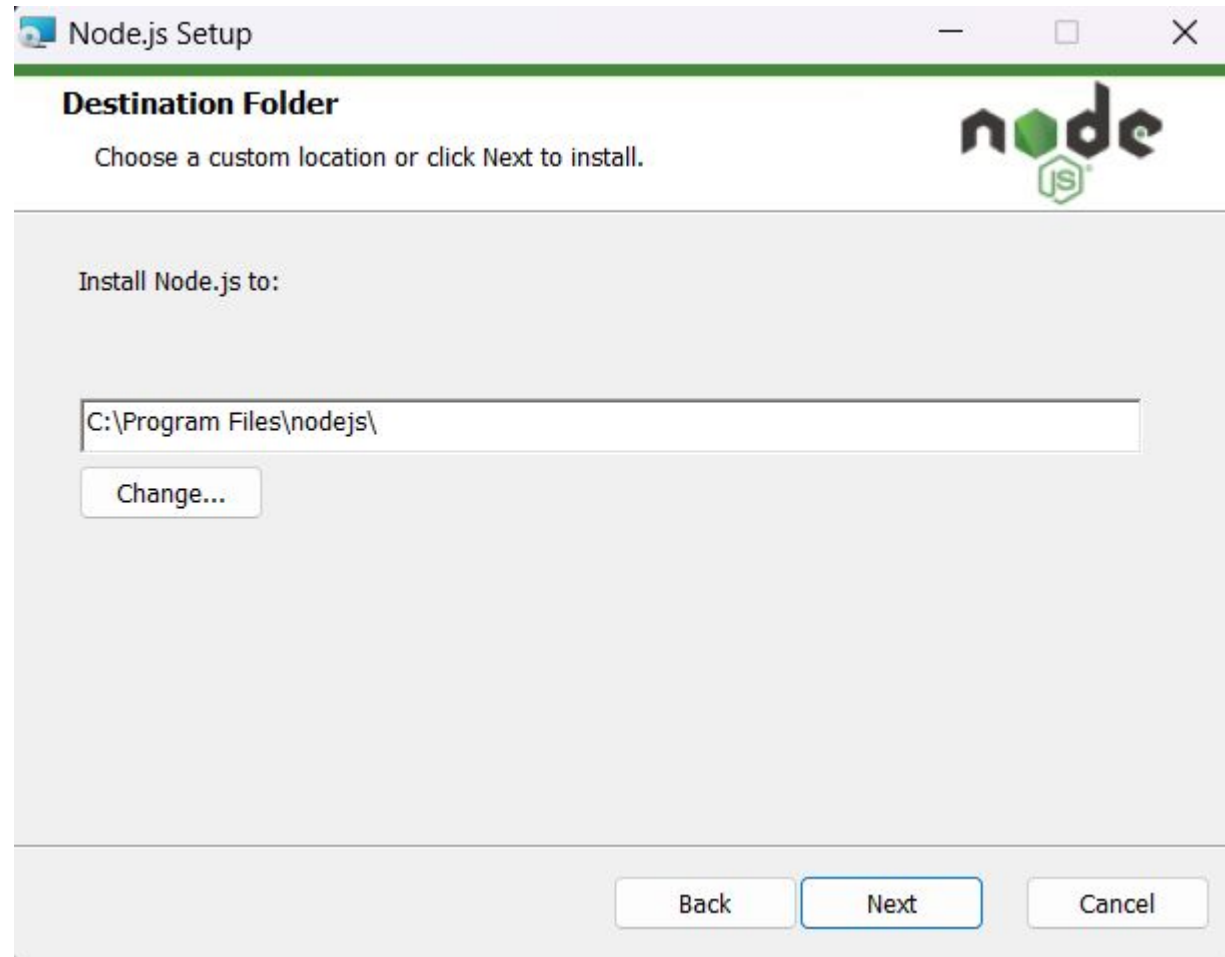
# Running React

- Download and Install - Node.js — Download Node.js®

**For linux**

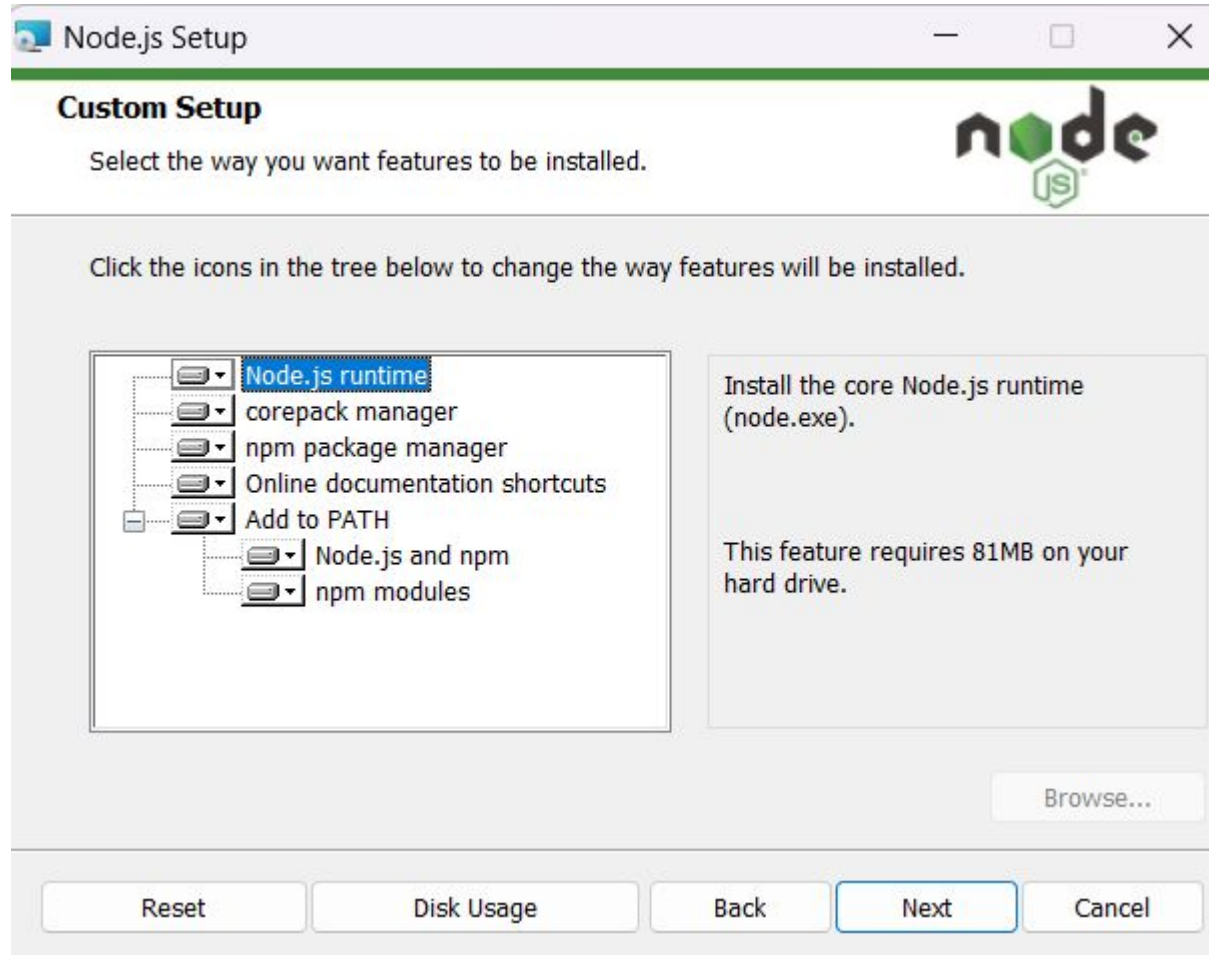
`sudo apt update`

`sudo apt install nodejs npm`



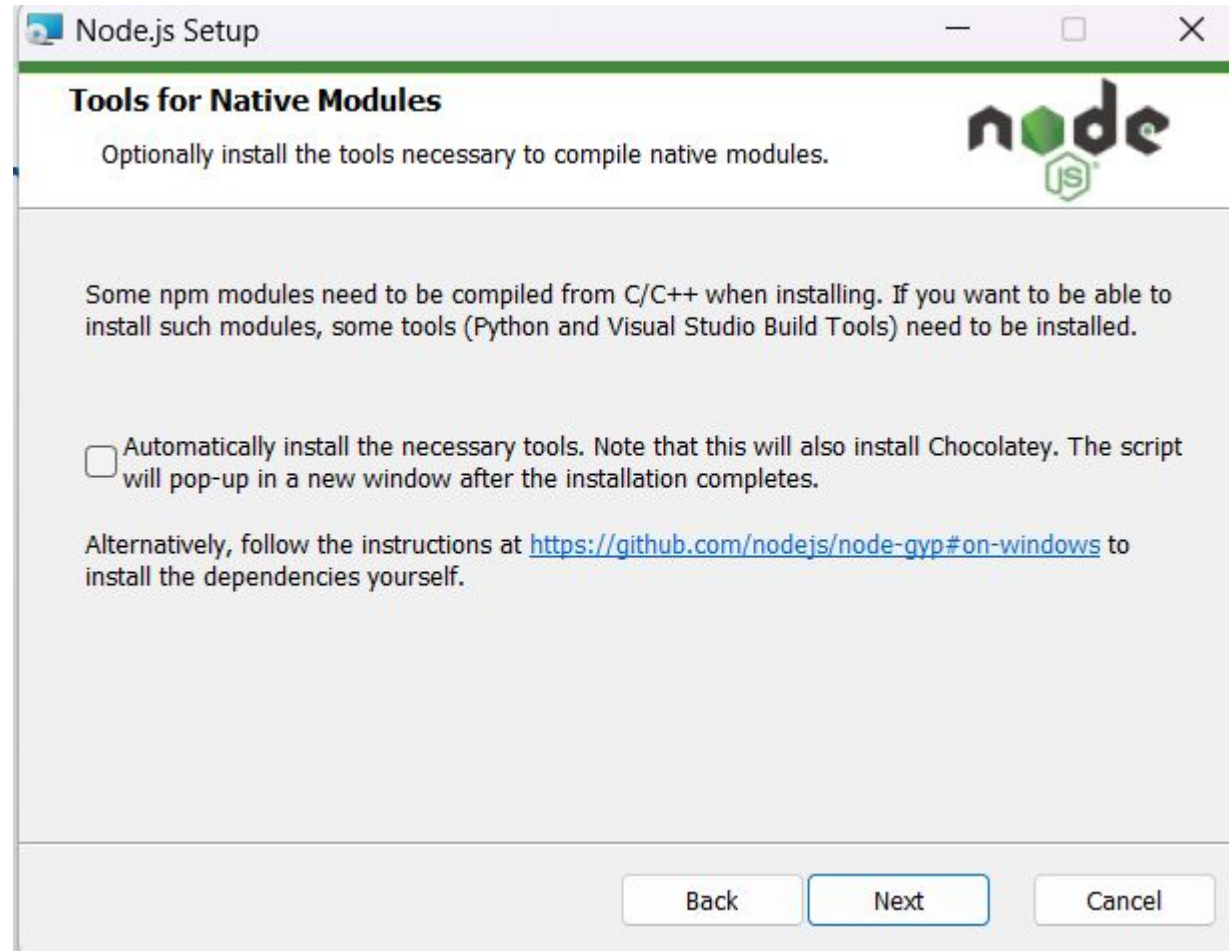
# Running React

- Download and Install - Node.js — Download Node.js®



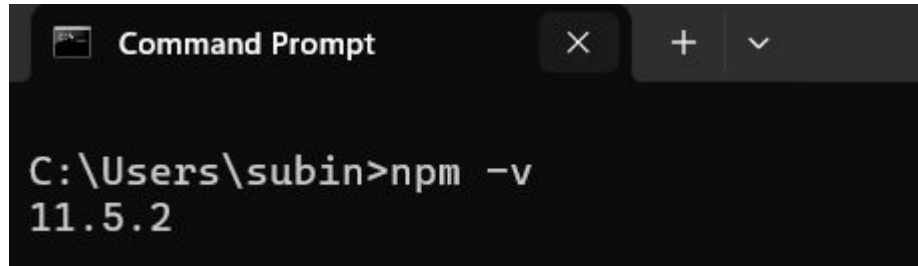
# Running React

- Download and Install - Node.js — Download Node.js®



# Running React

- Download and Install - Node.js — Download Node.js®



```
Command Prompt
C:\Users\subin>npm -v
11.5.2
```

- Verify after installation from VS code Terminal

```
node -v
npm -v
```

| Term    | Simple Meaning                                                      |
|---------|---------------------------------------------------------------------|
| Node.js | Runs JavaScript <b>outside the browser</b> (usually on the server). |
| npm     | Downloads and manages JavaScript libraries/packages.                |



# Running React

- Download and Install - Node.js — Download Node.js®

```
C:\Users\subin>D:  
D:\>cd react  
D:\react>
```

# Running React

- Download and Install - Node.js — Download Node.js®

```
D:\react>npm create vite@latest myapp -- --template react
Need to install the following packages:
create-vite@7.0.3
Ok to proceed? (y) y

> npx
> create-vite myapp --template react

|
o  Scaffold project in D:\react\myapp...
|
-  Done. Now run:

  cd myapp
  npm install
  npm run dev
```

# Running React

- Download and Install - Node.js — Download Node.js®

```
D:\react>cd myapp  
  
D:\react\myapp>npm install  
  
added 197 packages, and audited 198 packages in 25s  
  
33 packages are looking for funding  
  run `npm fund` for details  
  
found 0 vulnerabilities
```

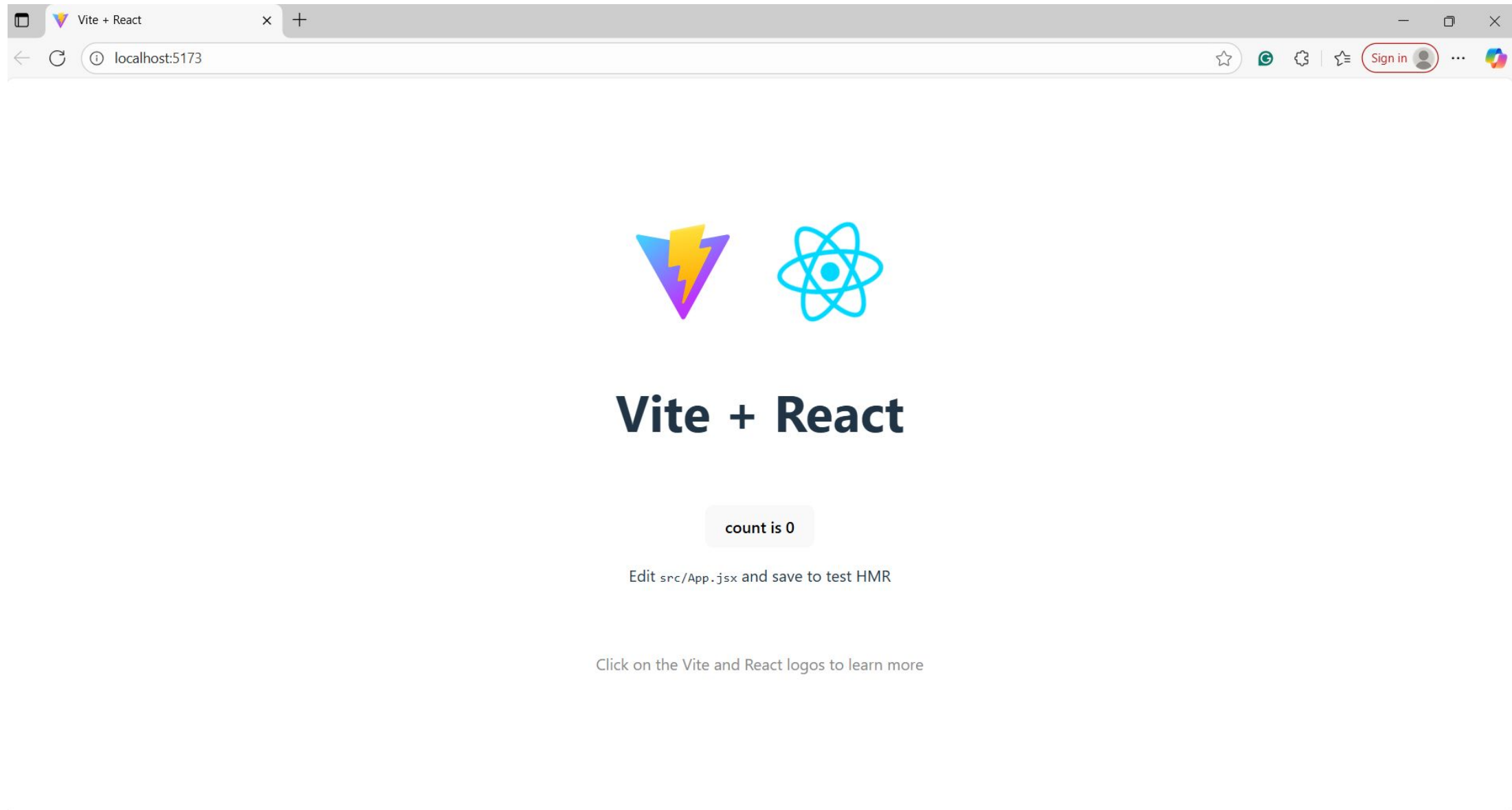
npm install  
npm run dev

# Running React

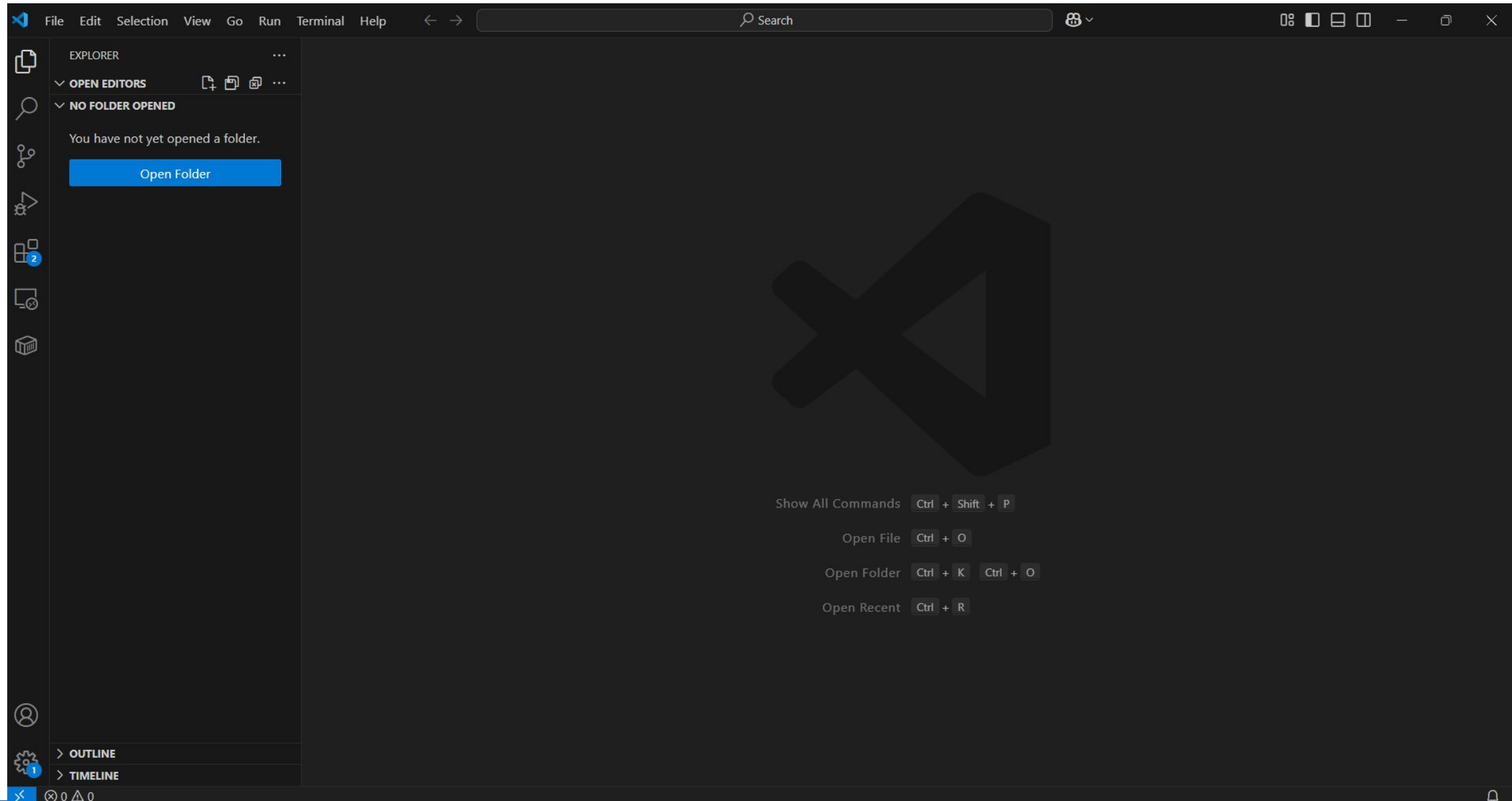
- Download and Install - Node.js — Download Node.js®

```
D:\react\myapp>npm run dev  
  
> myapp@0.0.0 dev  
> vite  
  
VITE v7.0.6 ready in 2744 ms  
  
→ Local:   http://localhost:5173/  
→ Network: use --host to expose  
→ press h + enter to show help  
|
```

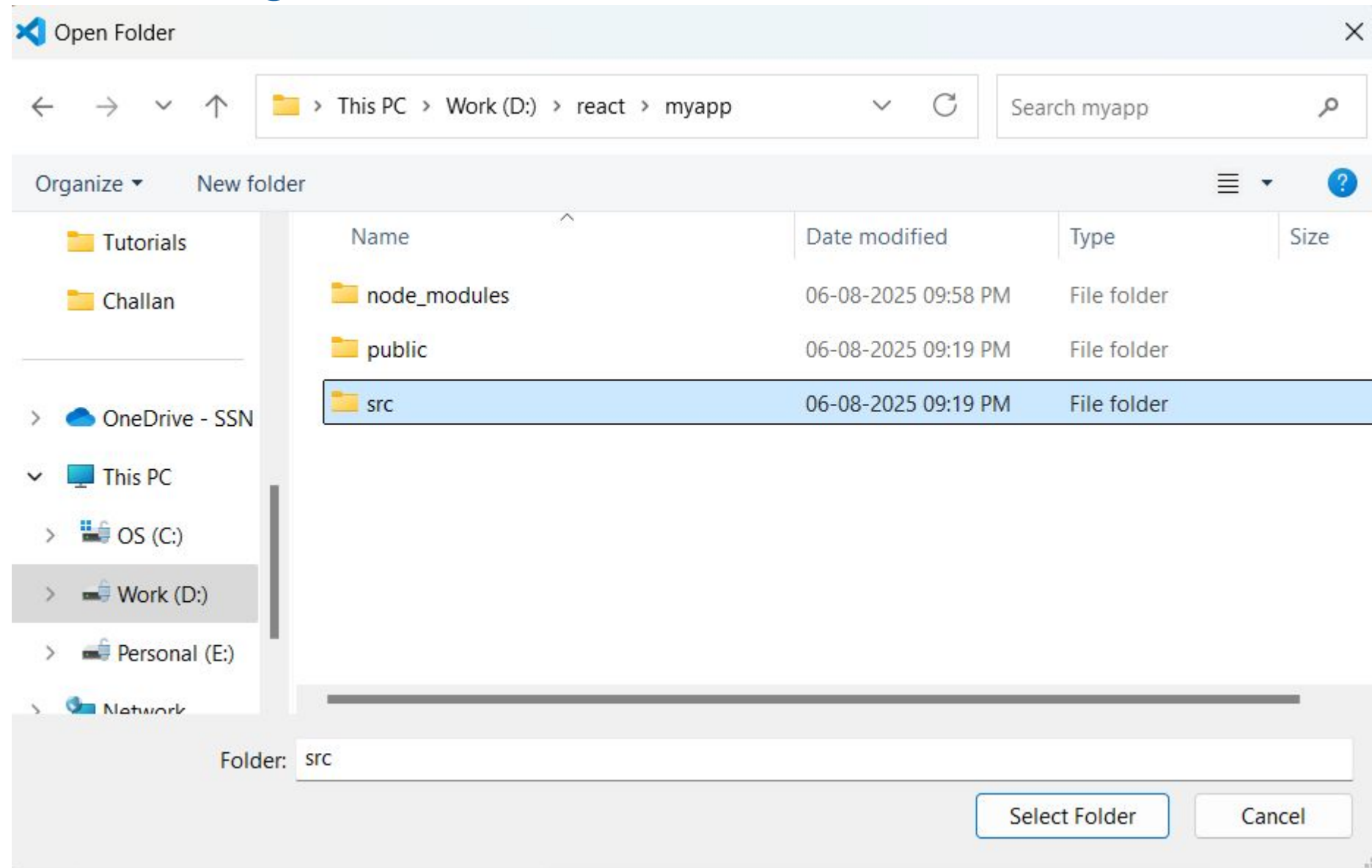
# Running React



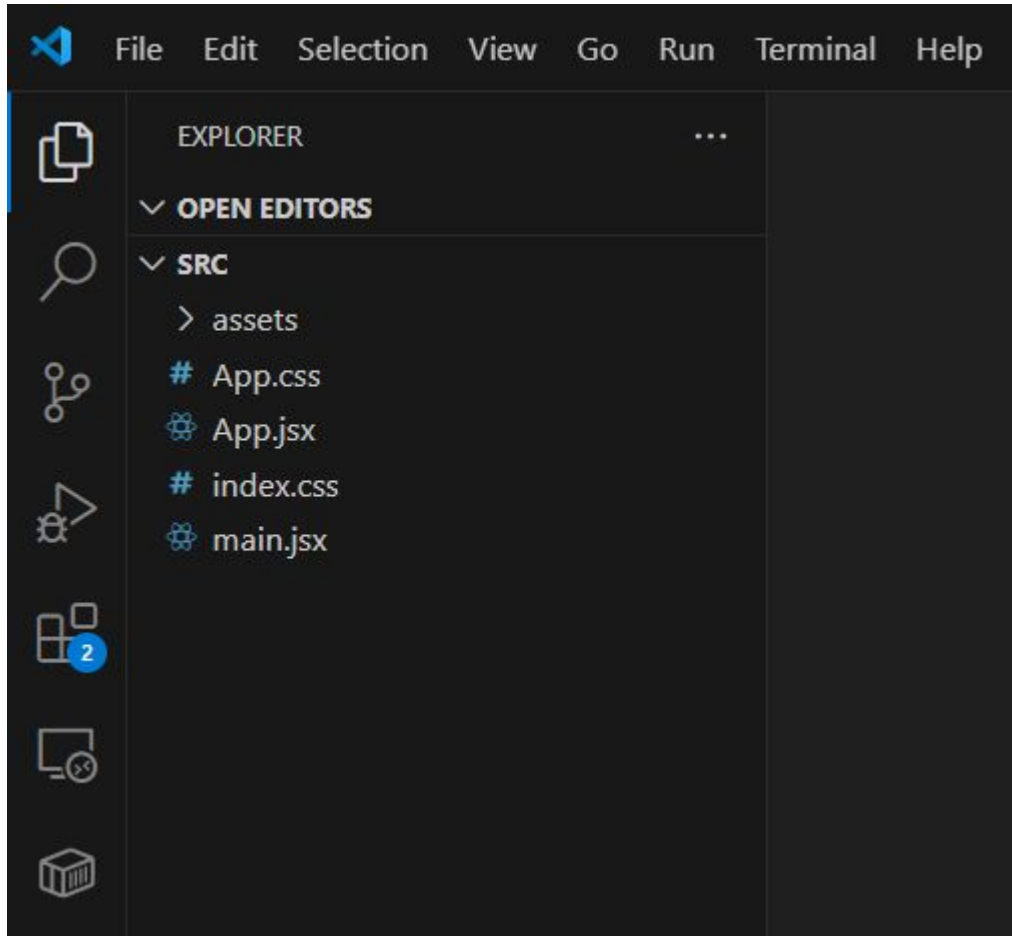
# Running React



# Running React

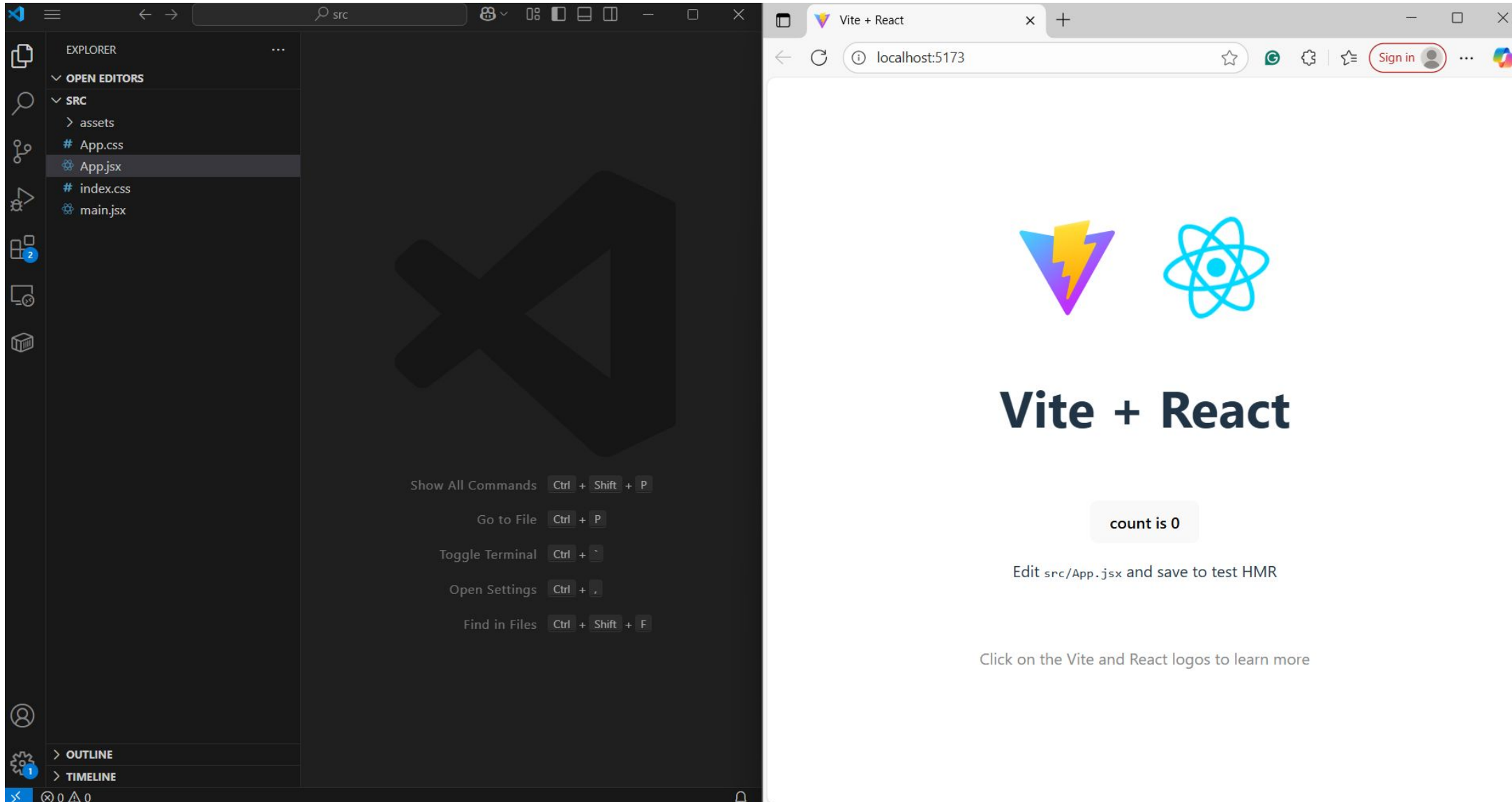


# Running React





# Running React



# Running React

```
App.jsx  X
App.jsx > App
1  import { useState } from 'react'
2  import reactLogo from './assets/react.svg'
3  import viteLogo from '/vite.svg'
4  import './App.css'
5
6  function App() {
7    const [count, setCount] = useState(0)
8
9    return (
10     <>
11       <div>
12         <a href="https://vite.dev" target="_blank">
13           <img src={viteLogo} className="logo" alt="Vite logo" />
14         </a>
15         <a href="https://react.dev" target="_blank">
16           <img src={reactLogo} className="logo react" alt="React logo" />
17         </a>
18       </div>
19       <h1>Vite + React</h1>
20       <div className="card">
21         <button onClick={() => setCount((count) => count + 1)}>
22           count is {count}
23         </button>
24         <p>
25           Edit <code>src/App.jsx</code> and save to test HMR
26         </p>
27       </div>
28       <p className="read-the-docs">
29         Click on the Vite and React logos to learn more
30       </p>
31     </>
32   )
33 }
34
35 export default App
36
```

# Running React

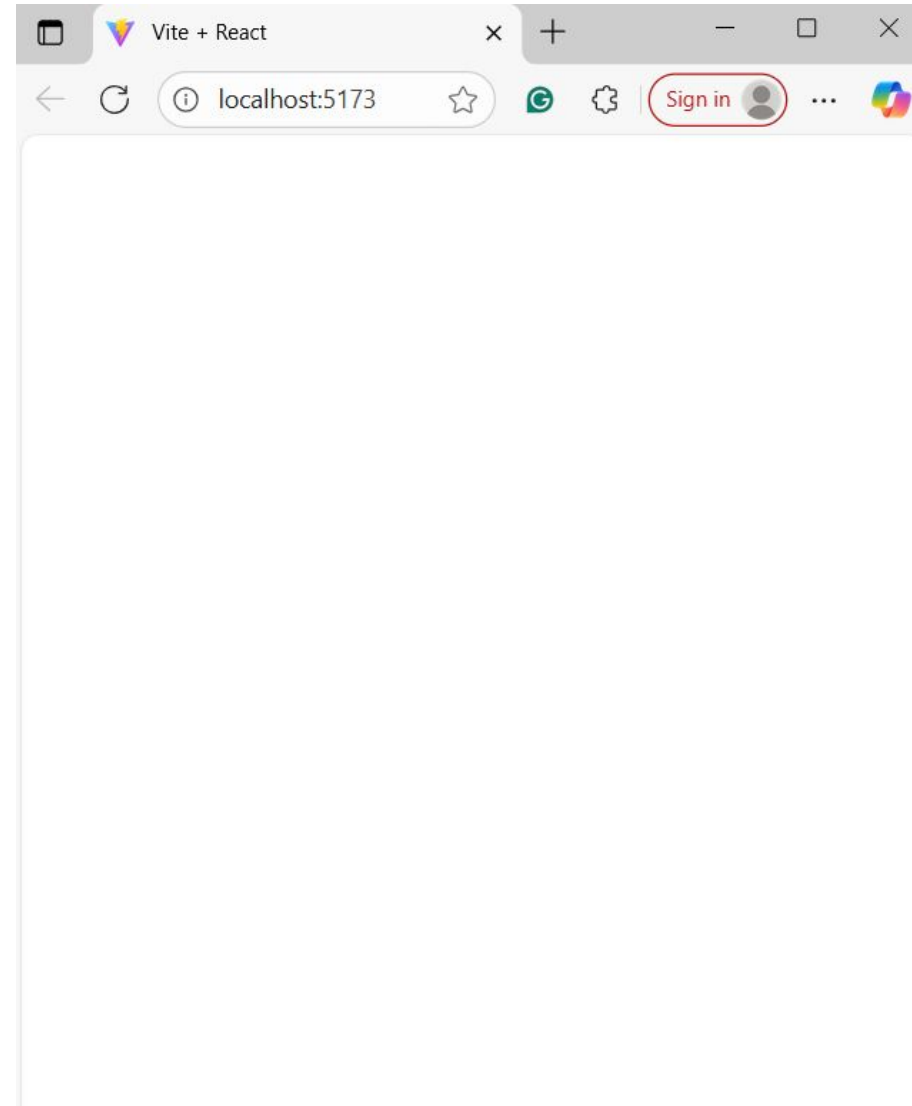
```
main.jsx ×  
main.jsx  
1  import { StrictMode } from 'react'  
2  import { createRoot } from 'react-dom/client'  
3  import './index.css'  
4  import App from './App.jsx'  
5  
6  createRoot(document.getElementById('root')).render(  
7    <StrictMode>  
8      <App />  
9    </StrictMode>,  
10 )
```

# Running React

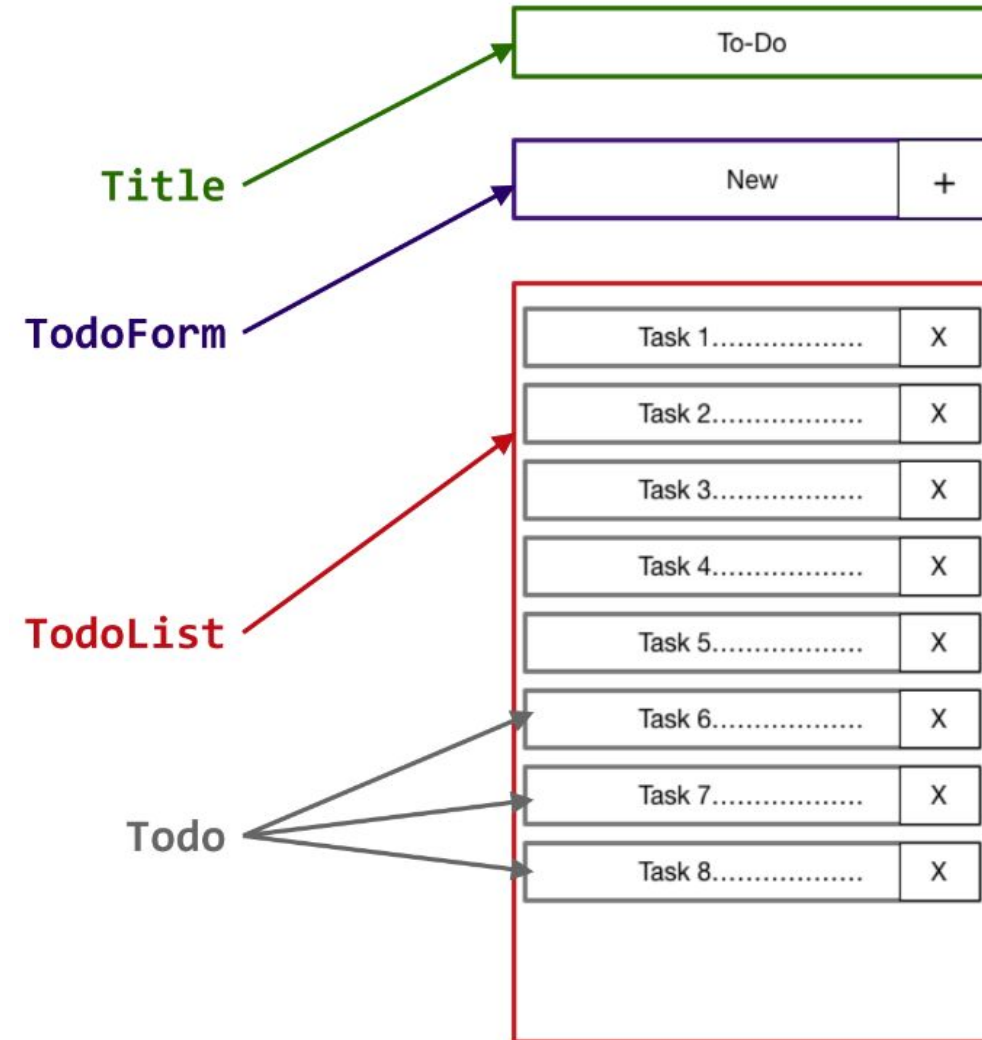
```
App.jsx ×  
App.jsx > ...  
1  import './App.css'  
2  
3  function App() {  
4  
5    return (  
6      <>  
7      </>  
8    )  
9  }  
10  
11  export default App
```

# Running React

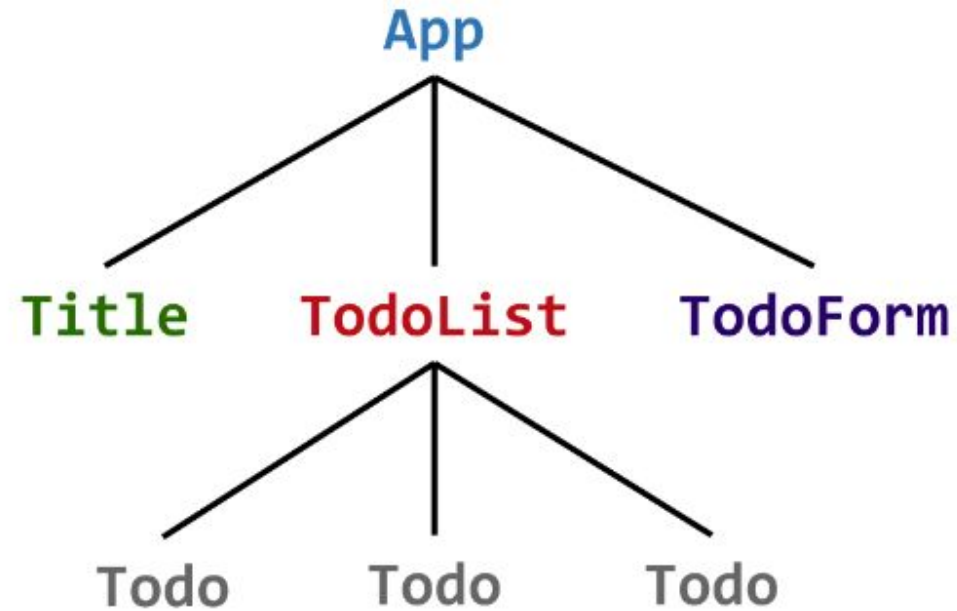
```
App.jsx ×  
App.jsx > ...  
1  import './App.css'  
2  
3  ✓ function App() {  
4  
5  ✓   return (  
6      <>  
7      </>  
8    )  
9  }  
10  
11  export default App
```



# React Component – Todo list

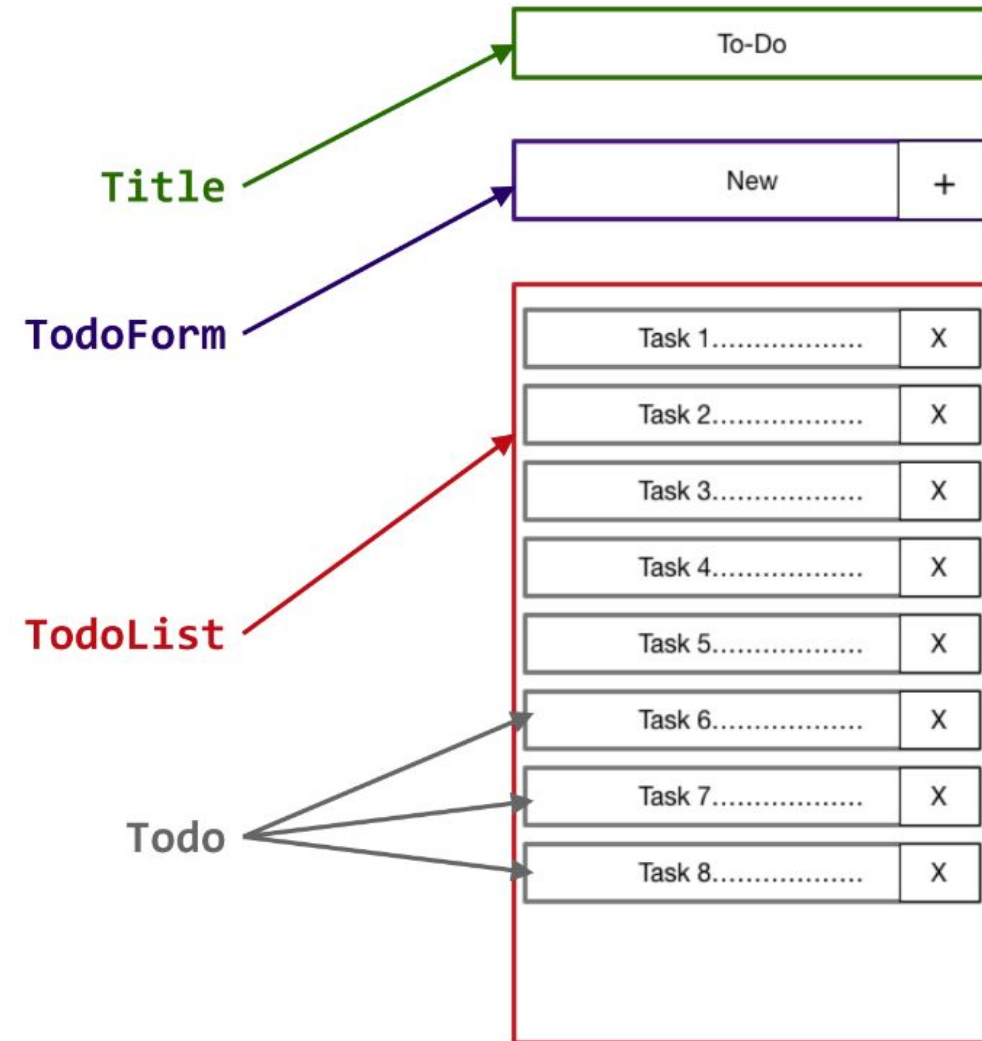


# React Component Hierarchy – Todo list



```
function App() {
  return (
    <Title />
  );
}
```

```
function TodoList() {
  return (
    <>
      <TodoForm />
      <TodoItem />
      <TodoItem />
    </>
  );
}
```



# React Props

## 1. Props?



# React Props

## 1. Props

- Properties
- Object

# React Props

## 1. Props

- Properties
- Object
- Function Arguments – Javascript

```
//Welcome Functional Component
function Welcome(props) {
  return <h1>Hello, {props.name} </h1> ;
}
```

# React Props

## 1. Props

- **Properties**
- **Object**
- **Function Arguments – Javascript**
- **Attributes – HTML**

```
//Welcome Functional Component
function Welcome(props) {
  return <h1>Hello, {props.name} </h1> ;
}
```

# React Props

## 1. Props

- Properties
- Object
- Function Arguments – Javascript
  - Syntax: `function identifier(props) {function body};`
- Attributes – HTML

```
//Welcome Functional Component
function Welcome(props) {
  return <h1>Hello, {props.name} </h1> ;
}
```

```
function App() {
  //Declaritive Use of Welcome Functional Component
  return (
    <>
      <Welcome name = "Subin"/>
    </>
  )
}

export default App
```

# React Props

## 1. Props

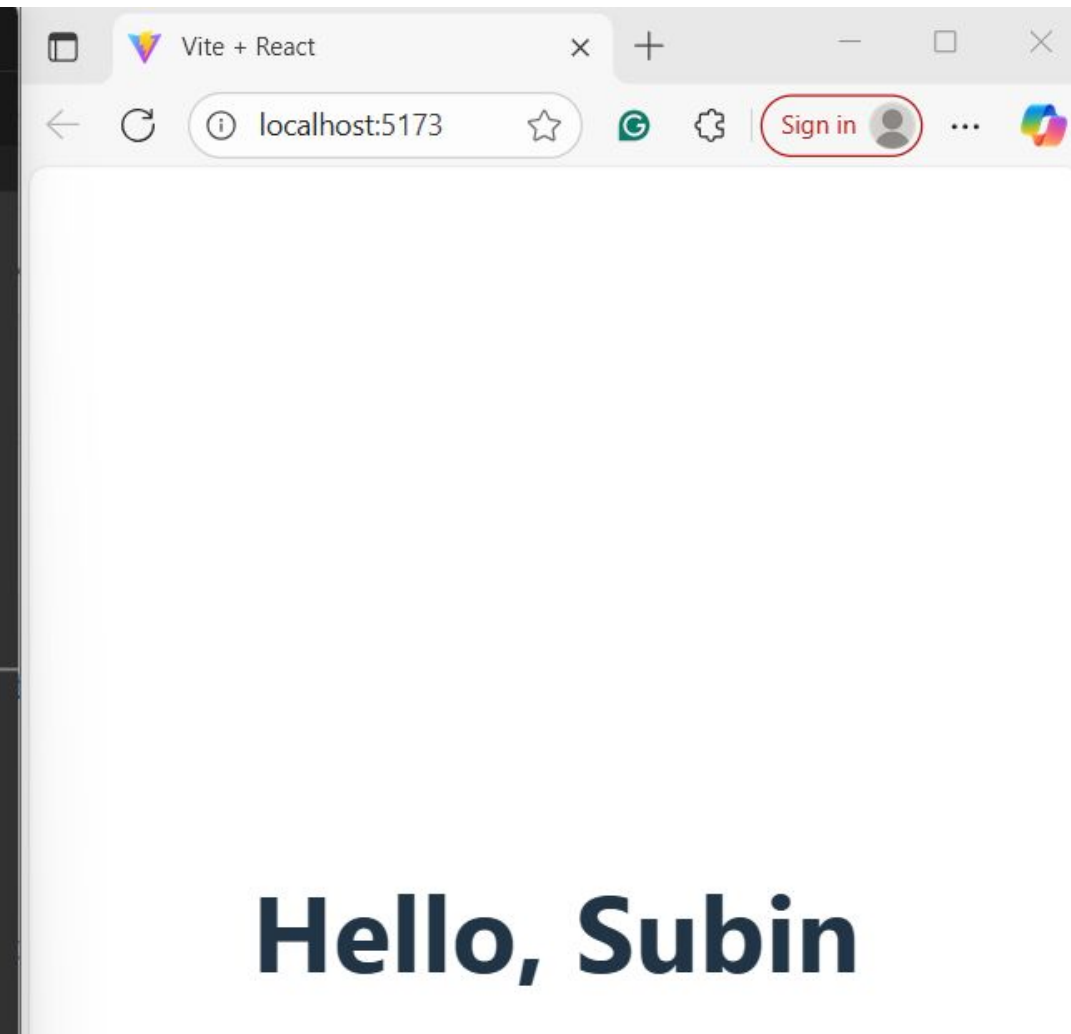
- Properties
- Object
- Function Arguments – Javascript
  - Syntax: `function identifier(props) {function body}`
- Attributes – HTML
  - Syntax: `<start_tag attribute="value">`

```
function App() {  
  //Declaritive Use of Welcome Functional Component  
  return (  
    <>  
    <welcome name = "Subin"/>  
    </>  
  )  
}  
  
export default App
```

```
//Welcome Functional Component  
function Welcome(props) {  
  return <h1>Hello, {props.name} </h1> ;  
}
```

# React Props

```
App.jsx
App.jsx > ...
1  import './App.css'
2  import React from 'react';
3
4  //Welcome Functional Component
5  function Welcome(props) {
6    return <h1>Hello, {props.name} </h1> ;
7  }
8
9  function App() {
10   //Declaritive Use of Welcome Functional Component
11   return (
12     <>
13       <Welcome name = "Subin"/>
14     </>
15   )
16 }
17
18 export default App
19 |
```



# React Props

```
react_search_filter_app/  
├── package.json  
├── vite.config.js  
├── index.html  
└── src/  
    ├── main.jsx  
    ├── App.jsx  
    ├── App.css  
    └── index.css
```

# React Props

```
react_props_app/  
├── package.json  
├── vite.config.js  
├── index.html  
└── src/  
    ├── main.jsx  
    ├── App.jsx  
    ├── App.css  
    └── index.css
```

## **package.json**

Defines the project.

Lists required libraries.

Defines commands such as  
npm run dev.



# React Props

```
react_props_app/  
├── package.json  
├── vite.config.js  
├── index.html  
└── src/  
    ├── main.jsx  
    ├── App.jsx  
    ├── App.css  
    └── index.css
```

## **vite.config.js**

Configures Vite.

Enables React support  
through the React plugin.

# React Props

```
react_props_app/  
├── package.json  
├── vite.config.js  
├── index.html  
└── src/  
    ├── main.jsx  
    ├── App.jsx  
    ├── App.css  
    └── index.css
```

## **index.html**

The only HTML page loaded by the browser.

Contains the root `<div>` where React is attached.

# React Props

```
react_props_app/  
├── package.json  
├── vite.config.js  
├── index.html  
└── src/  
    ├── main.jsx  
    ├── App.jsx  
    ├── App.css  
    └── index.css
```

## **src/main.jsx**

Entry point of the React application.  
Connects React with index.html.  
Renders the App component.

# React Props

```
react_props_app/  
├── package.json  
├── vite.config.js  
├── index.html  
└── src/  
    ├── main.jsx  
    ├── App.jsx  
    ├── App.css  
    └── index.css
```

**src/app.jsx**  
all the functions

# React Props

```
react_props_app/  
├── package.json  
├── vite.config.js  
├── index.html  
└── src/  
    ├── main.jsx  
    ├── App.jsx  
    ├── App.css  
    └── index.css
```

## **src/App.css**

Provides component-level styling for the search and product table UI.

## **src/index.css**

Provides global styles applied to the entire React application.

# References

1. **Paul J Deitel, Harvey M Deitel, and Abbey Deitel, "Internet and the world wide web: How to program," Pearson, Fifth Edition, 2012.**
2. **Robin Wieruch, "The Road to React: The ReactJS with Hooks", Lean Publishing, 2023 Edition.**

THANK YOU

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